

Encyclopedia

This entry analyses the origin of the word encyclopedia, the formal determinations of the encyclopedic genera, its more relevant modalities, both in its general form and in its philosophical dimension. It also includes a short reference to the more significant historical moments of encyclopedism as well as to the actual developments of encyclopedia, namely electronic and online encyclopedias, hypertext and internet as its actual realization.

The word

The word comes from Hellenistic expression *eu kuklios paideia* meaning the “perfect cycle of studies”. It is taken again in Roman culture appearing namely at the dedicace to the emperor Titus in the *Historia Naturalis* by Plinius.

In modern times, the word was first used in English by Sir Thomas Elyot (*Boke of the Governour*, 1531) and in French by Rabelais (*Pantagruel*, 1532).

There are many works which may be retroactively considered as belonging to encyclopedic genera although they had not that designation in their title (e.g. Vincent de Beauvais *Speculum Majus*, c.1260). In fact, it is only at the XVII century that the word encyclopedia comes close to its actual meaning. It is also then that encyclopedia becomes a true *genera* with its identity and characteristic procedures.

Encyclopedic genera

Encyclopedia aims to be the **exposition of all knowledge conquered by mankind**. This vertigo towards exhaustivity can lead encyclopedia to a teratological dimension as

in the case of the immense Chinese encyclopedias. However, in occidental world, encyclopedia is an economic work combining exhaustivity with selectivity.

Encyclopedia is a **historical production** reflecting a specific cultural and scientific situation. Never complete, always condemned to be surpassed by knowledge progress, it demands constant actualization. And actualization does not consist of only including new entries according to recent developments (e.g. bioethics) but also of diminishing the weight (e.g. flogistus) and of raising the importance of others (e.g. globalization).

Encyclopedia is **not a dictionary**. Dictionaries aspire to be a full, consistent codification of language. On the contrary, encyclopedia is semantically opened referring the things and events which are to be spoken, that is, known. Although many have been designated as dictionaries, encyclopedia is never interested in words but in what words mean and refer – the world behind the words.

Like dictionaries, encyclopedia is a **discontinuous text** made of independent *entries* which never present well-defined borders. Each entry opens (explicitly or implicitly) to other entries which, in turn, open to others in such a way that each entry is virtually connected with all others. Thus, encyclopedia is not so much the reunion of all knowledge in one closed place but the circulation (*navigation*) throughout the vast *net* of its multidimensional elements.

So, the material objectivity of encyclopedia has an **unlimited and combinatory condition**. Behind the additive synthesis of all its entries, encyclopedia points to the exhaustion of all the possible combinations of its entries which can be articulated in an undetermined number of combinations – a kind of combinatory without rule.

This situation has important consequences on the **reading regime** of encyclopedia. The readers of an encyclopedia are offered the possibility of making their own journey of

reading according to their interests and preferences (*surfing*). It is the reader (*navigator*) who selects, by successive choice, which semantic field he will read further on. In fact, encyclopedia promotes that possibility, proposing several resources which invite the reader to actualize one of the many possibilities it offers (indexes, thesaurus, internal references, cross-connections, articulations, instantaneous electronic *links* which today characterize hypertext and internet as its extension).

Encyclopedia extensively uses **non-linear materials** such as maps, drawings, illustrations, statistic schemata, plans and tables of all types (see the 11 complementary volumes of tables published by Diderot or the wide number of image resources in electronic encyclopedia, hypertext and internet). That is to say, encyclopedia points to the semantic exploration of the diagrammatic resources of language putting them at the service of the iconic, imagetic, cartographic description of the world.

Encyclopaedia is a **collective work**. Some were made by only one author (e.g. Pierre Bayle's *Dictionnaire Linguistique et Critique* (1647-1706). However, encyclopedia inevitably supposes the collaboration of different competencies: celebrated scientists, appointed experts, identified authors, anonymous, unidentified, even unknown collaborators (as in today's *Wikipedia*).

Implicitly or explicitly, encyclopedia is always an **ordered, organized structure**. It is that ordered structure that determines the quantity and quality of entries, the inclusion or exclusion of certain topics, the relative importance of some entries towards others. That does not mean that encyclopedia should be endowed with a unique, constraining point of view. What it means is that encyclopedia has an undeniable synthetic vocation.

Encyclopedia has always a **prospective role**. By promoting terminological unification, local systematizations, interdisciplinary connections, encyclopedia favors the

establishment of bridges from one domain to another. By synthesizing the already known, encyclopedia constitutes a kind of artificial prosthesis which liberates natural memory for what really matters – the unknown. It intends to reduce the opposition between memory and invention. That is why encyclopedia is such a strong anti-cartesian device

Modalities

Encyclopedia is essentially **generalist** offering an exhaustive yet concise exposition of the empirical and scientific knowledge acquired by mankind until a determined moment. They may be disciplinary (based on a classification of sciences); alphabetic (operating by definition of a huge set of concepts alphabetically ordered); thematic (constituted by large entries covering a set of selected subjects); mix, aiming to safeguard the synthetic benefits of disciplinary and thematic encyclopedias with practical advantages of alphabetic ordering. However, face to the development of scientific knowledge and the growing of cultural complexity encyclopedias were forced mostly at the XIX century to become **specialized**, focused on one human activity or knowledge domain. **Philosophical encyclopedias** are specialized encyclopedias aiming to cover philosophical tradition. Nevertheless, they may be considered to coincide with the very philosophical project. The cases of Leibniz and of Neurath are exemplar. For both it is through encyclopedia that philosophy reaches its aim of a unified knowledge of the world. It is true that Leibniz did not provide any completed encyclopedia. However he formulated with extreme rigor the theoretical problems concerned with such enterprise. And he clearly understood the need and heuristic significance of constructing in parallel encyclopedia and *characteristica universalis*. If the construction of the scientific language supposes the analysis of fundamental concepts and their definitions, that is, encyclopedia, inversely, the symbolic transposition of the cognitive

conceptual contents in the linguistic system already constituted, allows the discovery of new virtual meanings and thus favors encyclopedia. As for Neurath, he refused both the need of a foundationalist basis for encyclopedia and of its unique, total, *a priori* systematic organization. In doing so, he fully identified encyclopedia as the “model” of human knowledge and as the main task for unity of science. A unity which, like encyclopedia itself, should be plural, heterogeneous, always provisory, incomplete, but open to further cooperative contributions.

Historical Summary

From Plinius’ *Historia Naturalis* to *Etimologies* by Isidorus of Sevilla or *Eruditionis Didascalicae* by Huges de St. Victor it is possible to speak of a medieval and even Roman encyclopedia. However, in precise sense, encyclopedia is a product of the **XVII century**. Face to the intense scientific activity, encyclopedia appears as the *topos* where new forms of disciplinary and systematic arrangement of knowledge and new schemes of intelligibility are intended (Bacon’s *Instauratio Magna*, Alsted’s *Combinatoria*, Comenius’ *Pansofia*, Leibniz’ *Mathesis Universalis*).

XVIII century is the golden age of encyclopedism. Projects became effective realizations. Having reached its classical form, science offer itself the luxury of its own monumentality. Encyclopedia will play the role of medieval cathedral, a monument in which men face what is more important: not anymore God but the created World. The major realization is undoubtedly the *Encyclopédie ou Dictionnaire Raisonné des Sciences, des Arts et des Métiers*, (1751-1765) by Diderot and D’Alembert, a kind of laboratory where it is possible to grasp all the virtues and difficulties of encyclopedic project. Diderot and D’Alembert were able to assemble many of the most eminent intellectuals of the century as well as humble artisans of all areas of human activity.

They were able to construct a kind of a cartographic overview, not of the secrets of the world (of which God was up till now the first and final cause) but of the heterogeneous, contingent, precarious achievements of human knowledge. The illuminist encyclopedia of Diderot and D'Alembert is in fact a singular moment of the history of culture in which, for the first time, a desacralized world was envisaged, that is a reality with its own autonomy and intelligibility was offered to the human intelligence and industry as the background for an infinite cognitive conquest.

At the **XIX century**, two divergent lines of encyclopedism are put forward. Under the positivist regime proposed by Auguste Comte, encyclopedia becomes the instrument for the coordination of particular sciences. It must consider the achievements of each discipline and point to the logical law of their development. (e.g. *Grande Encyclopédie* by Berthelot and Dreyfus, 1886-1903). By other side, German romantic encyclopedism aims to construct a “total science” on the basis of the deep analogies among the particular sciences (Novalis's *Fragmente*, 1802) or to expose their organic unity while necessary parts of the philosophical whole (Hegel's *Enzyklopädie der Philosophischen Wissenschaften im Grundrisse*, 1817).

At the first decades of the **XX century** and in the context of logical-positivism, Otto Neurath has the merit of having conceived the original plan and of having effectively took charge of the organization of the *International Encyclopedia of Unified Science* of which only two volumes were published in 1938-1939.

The **decades of 40 to 60** were characterized by a generalized disillusion on the very idea of encyclopedia. Above all in literature, names like Alberto Savinio, Georges Perec, Italo Calvino or Jorge Luis Borges come to explore the paradoxes of any totalizing attempt and to announce the end of any encyclopedic effort.

A surprising renewal will take place at the **decade of 70**. On the contrary of what could be expected, encyclopedia takes up a new integrative model intending to critically prevail over the huge fragmentation of scientific knowledge and information. The most innovative encyclopedias (namely *Encyclopedia Universalis* (1968-1975) and *Encyclopedia Einaudi* (1977-1984) put into practice thematic, interdisciplinary structures dealing with controversial matters, pioneering concepts, current debates, unexpected articulations.

Actual developments

In the last decades of the XX century another decisive development took place. The hypertext by Ted Nelson (1965) and the World Wide Web by Tim Bernes-Lee (1991) are directly inscribed in the history of encyclopedia. That is why, as soon as the necessary technical conditions were put forward, classical encyclopedia promptly adhered to its own informatization (*Britannica's* first CDRom and *online* editions appeared respectively at 1993 and 1994). As if electronic encyclopedia and further hypertext and internet come to realize the primordial aspirations of encyclopedia itself.

Of course, in hypertext – as the actual limit of encyclopedia – there are enormous difficulties, terrible noise, dangerous lack of cartographic references, serious deficiency of selectivity and credibility of contents, etc. But there are as well enormous advantages. Like encyclopedia, hypertext is a general technology of knowledge operating by a regime of multiple choices among always growing yet never fully actualized contents. And the access to all those possibilities is done by the simple activation of a mouse's click.

Privileged place of the unity of science, encyclopedia – and hypertext as its today's limit – is a powerful form of resistance against the effects of specialization. Encyclopedia – and hypertext as its today's limit – is the only attempt of unification of knowledge which is effectively realized, the only material realization of unity of science that condenses and presents to the eyes of everybody a large scope of materials which could never be confronted in any other way.

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See also: Collective Memory, Embodied Cognition, Enlightenment, Interdisciplinarity, Knowledge Society, Neurath's Unity of Science and the Encyclopedic Project.

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